Holly A. Zaharchuk

Postdoctoral researcher (she/her)
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Research interests

How does the brain respond to linguistic variation? I use a combination of neural (EEG, fMRI, TMS) and behavioral (eye-tracking, response time/accuracy) methods to study the perception, representation, and processing of diverse speech signals. I am particularly interested in the effects of multilingual experience on the language system.

Education

The Pennsylvania State University — *University Park, Pennsylvania*
Ph.D., Psychology and Language Science, Specialization in Cognitive and Affective Neuroscience — *2018 - 2024*
- Advisor: Dr. Janet G. van Hell
- Dissertation: *Cross-language transfer in voice onset time: A window into perceptual adaptation in brain and behavior*

The Pennsylvania State University — *University Park, Pennsylvania*
M.S., Psychology — *2018 - 2020*
- Advisor: Dr. Janet G. van Hell
- Master's thesis: *We "might could" revisit syntactic processing: Studying dialectal variation with event-related potentials*

The University of Chicago — *Chicago, Illinois*
B.A., Psychology, Minor in Linguistics — *2011 - 2015*
- Advisor: Dr. Daniel Casasanto
- Honors thesis: *Does linguistic experience change mental representations of pitch?*

Honors and awards

Doctoral Dissertation Research Improvement Grant (Ling-DDRI) — *The Pennsylvania State University*
NSF grant for doctoral research on human language (SBE-2234907) — *Feb 2023 - Feb 2025*

Partnerships for International Research and Education Fellowship (PIRE) — *The Pennsylvania State University*
NSF grant for international language research (OISE-1545900) — *May 2022 - Jun 2022*

Adele Miccio Memorial Travel Award — *The Pennsylvania State University*
Travel award for professional development — *Dec 2021*

University Graduate Fellowship — *The Pennsylvania State University*
Competitive five-year fellowship for research doctoral students (100 awarded per year University-wide) — *Aug 2018 - Aug 2024*

University Scholar Award — *The University of Chicago*
Four-year merit scholarship for undergraduate students — *Sep 2011 - Jun 2015*

Peer-reviewed publications

Zaharchuk, H. A., Olson, H. E., Mechtenburg, H., Phillips, M. C., & Myers, E. B. (in press). Measuring age-related differences in phonetic gradiency with the visual analogue scale and eye tracking. *Journal of the Acoustical Society of America*.

Zaharchuk, H. A., Walker, A. J., Miller, A. R., Fernandez, C. B., & Van Hell, J. G. (in press). Expecting a challenge: A behavioral and neurophysiological investigation of talker identity in cross-dialectal speech perception. *Journal of Laboratory Phonology*.

Miller, A. R., Jończyk, R., **Zaharchuk, H. A.**, & Van Hell, J. G. (2025). Unlocking second language novel metaphor processing: Behavioral and ERP insights from first and second-language English users. *Psychophysiology*, *62*(5), e70066.

Zaharchuk, H. A., & Karuza, E. A. (2021). Multilayer networks: An untapped tool for understanding bilingual neurocognition. *Brain and Language*, *220*, 104977.

Zaharchuk, H. A., Shevlin, A., & Van Hell, J. G. (2021). Are our brains more prescriptive than our mouths? Experience with dialectal variation in syntax differentially impacts ERPs and behavior. *Brain and Language*, *218*, 104949.

Lucero, C., **Zaharchuk, H.**, & Casasanto, D. (2014). Beat gestures facilitate speech production. In P. Bello, M. Guarini,

M. McShane, & B. Scassellati (Eds.), *Proceedings of the 36th Annual Meeting of the Cognitive Science Society*. Red Hook, NY: Curran Associates.

Manuscripts in progress

Luthra, S., Mechtenburg, H., Olson, H. E., **Zaharchuk, H. A.**, & Myers, E. B. (2025). Stimulation of right temporal cortex enhances talker typicality judgments [Manuscript submitted, under review]. *Journal of Cognitive Neuroscience*.

Zaharchuk, H. A., & Van Hell, J. G. (2025). The effects of exposure variability and similarity on talker-independent adaptation to Spanish-accented English [Manuscript submitted, under review]. *Journal of Memory and Language*.

Litcofsky, K., **Zaharchuk, H. A.**, & Van Hell, J. G. (2024). Cross-modality structural priming in dialogue is influenced by interlocutor accent [Manuscript submitted, in revision]. *Journal of Neurolinguistics*.

Invited talks

Zaharchuk, H. A. (2025). Bilingualism: Who, what, when, where, why, and how. Guest lecture, University of Connecticut, March 3.

Zaharchuk, H. A. (2023). Sociolinguistics in the lab and the field. Guest lecture, University at Buffalo, March 13.

Zaharchuk, H. A., Shevlin, A., & Van Hell, J. G. (2022a). Are our brains more prescriptive than our mouths? Experience with dialectal variation in syntax differentially impacts ERPs and behavior. Lecture series, Processing Variation (ProVAR), Federal University of Rio de Janeiro, February 14.

Zaharchuk, H. A., Shevlin, A., & Van Hell, J. G. (2022b). Experiencia con variación dialéctica en sintaxis afecta el procesamiento de lenguaje. Panel session, III Congreso Paraguayo de Lingüística Aplicada (CONPLA), Asunción, Paraguay, November 15-17.

Conference presentations

Luthra, S., Mechtenberg, H., Olson, H. E., **Zaharchuk, H. A.**, Feder, A., & Myers, E. B. (2025, September). Evaluating a causal role for right temporal cortex in accessing talker-specific phonetic signatures [Poster session]. 17th Annual Meeting of the Society for the Neurobiology of Language (SNL), Washington, DC.

Zaharchuk, H. A., Washington, P. N., & Myers, E. B. (2025, September). Hemispheric differences in speech perception: The role of right temporal regions in talker-specific learning [Poster session]. 17th Annual Meeting of the Society for the Neurobiology of Language (SNL), Washington, DC.

Zaharchuk, H. A., & Van Hell, J. G. (2024, November). The neurocognitive correlates of talker-specific adaptation to Spanish-accented speech [Poster session]. 65th Annual Meeting of the Psychonomic Society, New York City, New York.

Walker, A., Hampton, C., Wood, W., Miller, A. R., **Zaharchuk, H. A.**, & Van Hell, J. G. (2024, April). Bidialectal brains: Profiles of event-related potentials in a cross-dialectal listening task in Southern US English speakers [Paper presentation]. Joint meeting of Language Variation across the South V and the Southeastern Conference on Linguistics 91.

Walker, A., **Zaharchuk, H. A.**, Miller, A. R., & Van Hell, J. G. (2023, December). Your brain on accents: Profiles of event related potentials in cross-dialectal listening in the US English context [Poster session]. 185th Meeting of the Acoustical Society of America, Sydney, Australia.

Zaharchuk, H. A., & Van Hell, J. G. (2023, October). Investigating the roles of acoustic-phonetic and lexico-semantic processing in resolving perceptual ambiguity in Spanish-accented English [Poster session]. 15th Annual Meeting of the Society for the Neurobiology of Language (SNL), Marseille, France.

Zaharchuk, H. A., Lipski, J., & Van Hell, J. G. (2023, August). Mi casa es tu posá: Exploring the bilingual mental lexicon in speakers of Spanish and Palenquero [Poster session]. Architectures and Mechanisms for Language Processing Conference (AMLaP 29), Donostia-San Sebastián, Spain.

Zaharchuk, H. A., Medina, V., Paterno, S., Viswanathan, N., & Van Hell, J. G. (2023, March). Sentential context impacts bilingual speech-in-speech recognition [Poster session]. 36th Annual Conference on Human Sentence Processing (HSP), Pittsburgh, Pennsylvania.

Zaharchuk, H. A., Walker, A., & Van Hell, J. G. (2023, March). Tracking the time-course of cross-dialect comprehension with ERPs [Oral and poster sessions]. 30th Annual Meeting of the Cognitive Neuroscience Society (CNS), San Francisco, California.

Walker, A., **Zaharchuk, H. A.**, Fernandez, C. B., & Van Hell, J. G. (2021, August). The role of dialect expectation on

lexical processing: EEG evidence [Oral session]. 5th Variation and Language Processing Conference (VALP5), Copenhagen, Denmark.

Zaharchuk, H. A., Shevlin, A., & Van Hell, J. G. (2020, May). We "might could" revisit syntactic processing: Studying dialectal variation with event-related potentials [Poster session]. 27th Annual Meeting of the Cognitive Neuroscience Society (CNS), Boston, Massachusetts.

Zaharchuk, H. A., Shevlin, A., & Van Hell, J. G. (2019, November). Auditory comprehension of double versus single modal constructions in Mainstream American English listeners [Poster session]. 60th Annual Meeting of the Psychonomic Society, Montréal, Québec, Canada.

Zaharchuk, H. A. (2015, June). Does linguistic experience change mental representations of pitch? [Poster session]. University of Chicago Undergraduate Research Symposium, Chicago, Illinois.

Academic employment

Language and Brain Lab — *Storrs, Connecticut*

Postdoctoral researcher — *Aug 2024 - present*

- Principal investigator: Dr. Emily B. Myers
- Research area: neurobiology of speech perception in young adults, older adults, and persons with aphasia
- Responsibilities: design, implement, and analyze research projects investigating laterality, gradiency, and talker specificity in speech perception with fMRI, TMS, and eye-tracking (NIH 5R01DC013064-09)

Bilingualism and Linguistic Diversity Lab — *State College, Pennsylvania*

Graduate assistant — *Aug 2018 - Aug 2024*

- Principal investigator: Dr. Janet G. van Hell
- Research area: neurocognitive mechanisms of auditory language comprehension in bidialectal and bilingual listeners
- Responsibilities: design, implement, and analyze research projects investigating cross-dialectal communication with EEG (NSF BCS-2041264 and BCS-2041081)

Experience and Cognition Lab — *Chicago, Illinois*

Undergraduate research assistant — *Sep 2013 - Jun 2015*

- Principal investigator: Dr. Daniel Casasanto
- Research area: embodied cognition and linguistic relativity
- Responsibilities: collect, code, and analyze behavioral data

Teaching experience

Psychology of language — *The Pennsylvania State University*

Instructor — *Jan 2024 - May 2024*

- Seminar course for upper-level psychology and linguistics undergraduates on psycholinguistics, with a focus on language development, bilingualism and second language acquisition, and linguistic diversity
- Presented bi-weekly lectures to highlight key concepts and clarify models and methods; mentored students on interpreting empirical research through reflection papers, journal article presentations and discussions, and individual meetings; fostered student curiosity and creativity through final project proposal and poster presentation

Psychology of language — *The Pennsylvania State University*

Teaching assistant — *Jan 2021 - May 2021*

- Instructor: Dr. Katharine Donnelly Adams
- Seminar course covering auditory speech perception, visual word recognition, language representation, sentence comprehension and production, and language acquisition across the lifespan
- Presented guest lecture on linguistic variation and the relation between production and comprehension

Basic research methods in psychology — *The Pennsylvania State University*

Teaching assistant — *Aug 2020 - Dec 2020*

- Instructor: Dr. Nicholas Pearson
- Introduction to experimental design, analysis, and presentation
- Led weekly laboratory section to reinforce lecture concepts

What makes us human? — *The Pennsylvania State University*

Teaching assistant — *Jan 2020 - May 2020*

- Instructor: Dr. Daniel Weiss
- Senior seminar in comparative psychology

Neurological bases of human behavior — *The Pennsylvania State University*

Teaching assistant — *Aug 2019 - Dec 2019*

- Instructor: Dr. Stephen Wilson
- Introduction to neuroscience theories and methods
- Presented guest lecture on auditory system and neuroanatomy of language production and comprehension

Academic service

- Ad hoc reviewing
  - *Brain and Language*
  - *Cognitive, Affective, and Behavioral Neuroscience*
  - *Journal of Experimental Psychology: Learning, Memory, and Cognition*
  - *Journal of Speech, Language, and Hearing Research*
  - *Perspectives*
  - *Psychonomic Bulletin & Review*
  - *Scientific Reports*
- Workshops and tutorials
  - Leader: R Markdown for Graduate Students (2020) for 20 students, Creating HTML Reports with R Markdown (2020) for 10 students
  - Assistant: R Bootcamp led by Dr. Rick Gilmore for 50+ students
- R Markdown for Psychology Graduate Students resource

Outreach

- Developing a Community Engagement Training Program for researchers in collaboration with the NIH T32 Cognitive Neuroscience of Communication - Connecticut (CNC-CT) program
- Volunteered with the Auditory Brain Research Lab to provide free hearing screenings in English and Spanish at the 4th Multicultural Latin Festival (July 2025, Willimantic, CT)
- Mentored underrepresented and first-generation students applying for graduate programs in cognitive psychology as part of the Mentoring Aspiring Graduate students and building an Inclusive Community (MAGIC) program at UConn (2024-2025)

Skills

- Natural languages: English, Spanish (B2/Intermediate)
- Programming languages: R, MATLAB, Python, SQL
- Experiment programming: E-Prime, PsychoPy, Gorilla, LabVanced, Pavlovia
- Acoustic analysis: Praat

Affiliations

- Center for Language Science (Penn State)
- Cognitive Neuroscience Society
- Institute for the Brain and Cognitive Sciences (UConn)
- Psychonomic Society
- Society for the Neurobiology of Language
- Women in Cognitive Science